

ARTICLE



Improving econometric prediction by machine learning

Giovanni Cerulli 

CNR-IRCRES, Research Institute on Sustainable Economic Growth, National Research Council of Italy, Rome, Italy

ABSTRACT

We present a Machine Learning (ML) toolbox to predict targeted econometric outcomes improving prediction in two directions: (i) by cross-validated *optimal tuning*, (ii) by comparing/combining results from different learners (*meta-learning*). In predicting woman wage class based on her characteristics, we show that all our ML methods' predictions highly outperform standard multinomial logit ones, both in terms of mean accuracy and its standard deviation. In particular, we set out that a regularized multinomial regression obtains an average prediction accuracy almost 60% larger than that of an unregularized one. Finally, as different learners may behave differently, we show that combining them into one ensemble learner proves to preserve good predictive accuracy lowering the variance more than stand-alone approaches.

KEYWORDS

Machine learning; ensemble methods; optimal prediction

JEL CLASSIFICATION

C53; C61; C63

I. Introduction

The quest for an objective science is in a surge. In economics, for example, some scholars have recently stressed the need for sounder credibility and fairness of empirical research (Angrist and Pischke 2010). Machine Learning (ML), a relatively new approach to data analysis, may help in this direction.

Placing itself in the intersection between statistics, computer science, and artificial intelligence, ML main objective is turning information into valuable knowledge by 'letting the data speak', limiting model's prior assumptions, and promoting a model-free philosophy. Relying on algorithms and computational techniques, ML targets Big Data and complexity reduction, although sometimes at expenses of results' interpretability (Mullainathan and Spiess 2017; Varian 2014).

ML has emerged as a new scientific paradigm within numerous sciences, but its use in economics and econometrics is still lagging behind. One shared belief among economists is that ML may be powerful for prediction, whereas less useful for inferential purposes (Athey 2019). Recently,

however, a new econometric literature is trying to bridge ML and causal inference through new ML-adapted methods able to tackle causal inference issues, such as treatment effect estimation with high dimensional data (Belloni, Chernozhukov, and Hansen 2014), heterogenous treatment effect estimation (Athey and Wager 2017), and optimal policy assignment (Athey and Imbens 2017).

Primarily focused on prediction, this paper can offer useful insights also to improve causal inference through, for example, a more accurate estimate of the so-called *selection equation*, generally estimated by (binomial or multinomial) probit or logit regressions to obtain propensity scores at individual levels. The selection analysis, the first step of any causal inference study, entails a 'prediction problem' so that the procedure proposed in this work is worthy whenever the selection-into-treatment generating process is not well approximated by parametric models such as probit or logit.¹

This paper presents a Machine Learning (ML) approach to predict targeted econometric outcomes improving prediction in two directions: (i)

CONTACT Giovanni Cerulli  giovanni.cerulli@ircres.cnr.it  CNR-IRCRES, Research Institute on Sustainable Economic Growth, National Research Council of Italy, Rome 00185, Italy

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¹The use of, let's say, a tree-based propensity score for estimating the selection equation opens up the problem of correctly estimating the standard error of the average treatment effect in the second-step (outcome) equation. What is the asymptotic distribution of the average treatment effect estimator when the first-step propensity score is estimated via a highly non-parametric procedure? This is still an open question that could forage a new stream of research in causal inference. A possible empirical solution could be the use of the bootstrap, although one should prove that bootstrap is correct in this context.

by cross-validated *optimal tuning*, (ii) by comparing/combining results from a large set of learners (*meta-learning*²).

The application presented in this paper focuses on classification, but for the sake of completeness, we briefly present also results for regression. Specifically, we aim at predicting the wage class (categorized as ‘low’, ‘medium’, and ‘high’) of an individual based on her characteristics. First, we show that all our ML methods predictions highly outperform standard multinomial logit ones, both in terms of mean accuracy and its standard deviation. This result is relevant, as the multinomial logit is a workhorse of econometric models for categorical outcomes, including its use in multi-valued propensity-score estimation. We show that all the estimated ML methods obtain a prediction accuracy higher than that of a standard multinomial, and that the regularized version of the multinomial is almost 60% more predictive.

As different learners may behave differently, we then show that combining them into one single ensemble learner proves to preserve good predictive accuracy lowering the variance more than stand-alone approaches. In regression settings, this can be done using a weighted average of single learners’ predictions, whereas in classification settings this can be achieved via a ‘majority vote’ ensemble rule.

More in general, by comparing ML methods, we argue that it is relevant to stress the role played not only by larger predictive accuracy but also by wider accuracy uncertainty.

The structure of the paper is as follows. [Section 2](#) presents the logic of the meta-learning architecture. [Section 3](#) illustrates our application and discusses the results. [Section 4](#) concludes the paper.

II. The meta-learning architecture

I define a learner L_j as a mapping from the set $[X, \theta, \lambda_j, f_j(\cdot)]$ to an outcome y , where X is the matrix of features, θ a vector of estimation parameters, λ_j a vector of tuning parameters, and $f_j(\cdot)$ an algorithm taking as inputs X , θ , and λ_j . It is common practice to use in applications a singleton $f_j(\cdot)$ for modelling and predicting

targeted outcomes, typically one member of the Generalized Linear Models (GLM) family (linear, probit or multinomial regressions are classical examples). GLM are, however, highly parametric and are not characterized by tuning parameters. Nonparametric models, such as local-kernel, nearest-neighbour, or tree-based regressions are on the contrary characterized by one or more hyperparameters λ_j which may be optimally chosen to minimize the so-called *test prediction error*, i.e. the out-of-sample predicting accuracy of the learner. In this direction, these methods can help improving prediction, although possibly at expenses of model results’ interpretability.

[Figure 1](#) presents the meta-learning architecture herein proposed. This framework is made of three linked learning processes: (i) the learning over the tuning parameter λ , (ii) the learning over the algorithm $f(\cdot)$, and (iii) the learning over new additional information. The departure is in point 1, from where we set off assuming the availability of a dataset $[X, y]$.

The first learning process aims at selecting the optimal tuning parameter(s) for a given algorithm $f_j(\cdot)$. ML scholars typically do it using K -fold cross-validation, a resampling approach estimating the out-of-sample performance of a learner by leaving one group of observations out of the estimation, and then using prediction over these left-out observations to measure predictive accuracy. This procedure is iteratively repeated for each fold, eventually obtaining K test-accuracy (or, equivalently, test error) measures over which taking the average and the standard deviation.

At the optimal λ_j , one can recover the largest possible prediction accuracy for the learner $f_j(\cdot)$. Further prediction improvements can be achieved only by learning from other learners, namely, by exploring other $f_j(\cdot)$, with $j = 1, \dots, M$ (where M is the number of learners at hand).

It is important to observe that the so-called *training error*, i.e. the in-sample predictive performance of a learner, is a misleading measure of the actual model fit as plagued by the *overfitting* phenomenon: it may be the case that the training error decreases monotonically with the tuning parameter even if the out-of-sample performance of the

²The main reference on the statistics of *meta-learning* can be found in Van der Laan and Rose (2011).

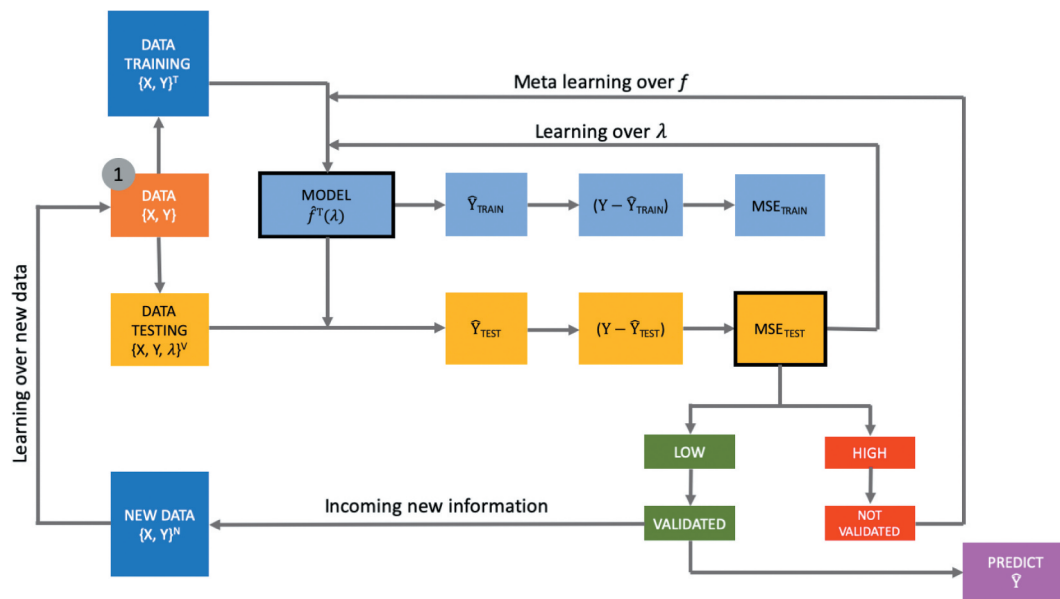


Figure 1. The meta-learning architecture.

learner is worsening. In Figure 1, it corresponds to the light blue sequence of boxes leading to the MSE_{TRAIN} which is in fact a dead-end node, as not informative for making correct decisions.

Conversely, the yellow sequence leads to the MSE_{TEST} , which is informative to take correct decisions about the predicting quality of the current learner. At this node, the analyst can compare the current MSE_{TEST} with a benchmark one (possibly, pre-fixed), and conclude whether to predict using the current learner, or explore alternative learners in the hope of increasing predictive performance. If the level of the current prediction error is too high, the meta-learning architecture would suggest to explore other learners.

In the ML literature, learning over learners is called *meta-learning*, and entails an exploration of the out-of-sample performance of alternative algorithms $f_j(\cdot)$ with the goal of identifying one behaving better than those already explored (Hastie, Tibshirani, and Friedman 2001). For each new $f_j(\cdot)$, the meta-learning architecture finds an optimal tuning parameter and a new estimated accuracy (along with its standard deviation). The analyst can either explore the entire bundle of alternatives and finally pick-up the best one, or

decide to select the first learner whose accuracy is larger than the benchmark. Alternatively, she can properly combine single learners' prediction to generate a unique prediction via some form of model averaging (*ensemble learning*). What might be the advantage of this procedure? As an average, an ensemble learner cannot provide the largest accuracy possible. However, as sums of i.i.d. random variables have smaller variance than the single addends, the benefit consists of a smaller predictive uncertainty (Zhou 2012; Escanciano, Jacho-Chvez, and Lewbel 2014).

The third final learning process concerns the availability of new information, via additional data collection. This induces a reiteration of the initial process whose final outcome can lead to choose a different algorithm and tuning parameter(s), depending on the nature of the incoming information.

The increase of prediction accuracy that this architecture may induce depends on the specific dataset employed. Depending on the nature of the data, one ML method may outperform all the others. Understating why this occurs is still an open issue in the ML literature, and for this reason the proposed architecture may come in handy: as the best method over all possible datasets does not

exist, it is reasonable to fit many algorithms to the same data so to singling out the most appropriate to use (or a combined ensemble of methods). This goes in the direction of providing prediction analysis with larger robustness.³

III. Application

We are interested in predicting the wage class of an individual based on her characteristics.⁴ We have data on 1,848 women from the U.S. National Longitudinal Survey of Young Women (NLSW) in 1988. Our target dependent variable, *wage_class*, is categorical with three classes of hourly wage: *low*, *medium*, and *high*. As individual features, we consider: *age*: age of the woman; *race*: race of the woman (white, black, other); *married*: married vs. non-married; *never_married*: whether or not never married; *grade*: grade obtained at school final exam; *south*: whether or not the woman comes from the South; *smsa*: whether she lives in SMSA; *c_city*: whether or not she lives in central city; *collgrad*: whether she is college graduated; *hours*: usual hours worked; *ttl_exp*: total work experience; *tenure*: job tenure in years; *industry*: type of industry; *occupation*: type of occupation.

Table 1 sets out the eight learners considered in this study. Except the Naïve Bayes, all the other learners present at least one tuning parameter over which we optimize. These learners are Boosting, Decision Tree, Naïve Bayes, Nearest-Neighbour, Neural Network, Random Forest, Regularized Multinomial, and Support Vector Machine.

Table 1. Learners and related tuning parameters.

Learner	λ_1	λ_2	λ_3
Boosting	Learning rate	Bootstraps	Tree-depth
Decision Tree	Leaves	–	–
Naïve Bayes	–	–	–
Nearest Neighbour	Neighbours	–	–
Neural Network	Hidden layers	Neurons	–
Random Forest	Splitting features	Bootstraps	Tree-depth
Regularized Multinomial	Penalization	–	–
Support Vector Machine	C	Γ	–

Figure 2 shows each learner's test and training accuracy as a function of the (main) tuning parameter and identifies the optimal tuning parameter as the one maximizing test accuracy. Focusing on the Decision Tree, for example, we clearly observe the overfitting produced by increasing the number of leaves (i.e. a monotonic increase of the training accuracy), which is optimal at a value of 7, where the test accuracy is maximized. A similar pattern can be observed for the Nearest-Neighbour, where the optimal numbers of neighbours is 17. In this case, the overfitting occurs as a function of $1/K$, with K the number of nearest neighbours.

As reference prediction accuracy, we consider the one obtained by a standard multinomial regression. The mean accuracy and its standard deviation are estimated using a 30-fold cross-validation procedure. The obtained accuracy is thus a good proxy of the out-of-sample predictive power of this approach. Observe that a multinomial model is not characterized by hyperparameters. For this model, as reported in Table 2, we obtain an out-of-sample prediction accuracy equal to 0.40 with a standard deviation of around 0.09, a huge one. By looking at the rest of the Table, we see that standard multinomial predictive performance, both in terms of mean and standard deviation accuracy, is the poorest vis-à-vis the ML methods considered in this study. Figure 3 sets out a visual summary of the results and provides useful insights on the predictive power of the considered classifiers. Some comments follow.

The Regularized Multinomial model provides the best accuracy of 63% which is the out-of-sample probability to correctly classify the wage class of a new out-of-sample woman. The worst classifier, other than the standard multinomial, in this dataset, is the Nearest Neighbour with an accuracy of 51%, although characterized by a relatively small variation.

Indeed, information on the accuracy's standard error is also relevant, as determining accuracy confidence interval. In this regard, the column with

³A challenging stream of research aims at understanding the relationship between data structure and ML models' prediction ability. We know so far that when data present a strong *inner ordering*, some methods tend to outperform others. For instance, for image recognition purposes, deep neural networks are surprisingly accurate compared to other classification algorithms. This has to do with the inner ordering of images, such a human faces. In particular, convolutional neural networks are highly suited for this task.

⁴All algorithms implementation and graphing have been programmed in Python 3.7, using the Stata/Python integrated interface available in Stata 16. All codes are available on request.

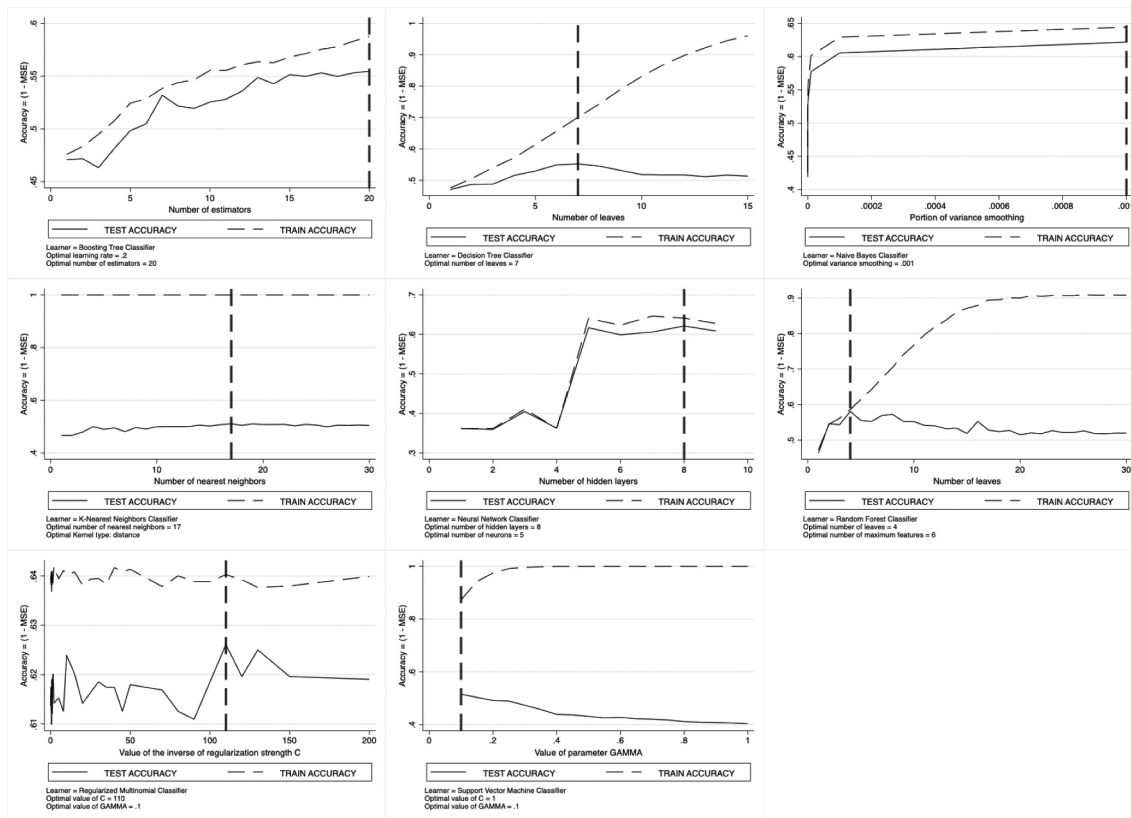


Figure 2. Learners' test and training accuracy as a function of the main tuning parameter.

Table 2. Learners' predictive accuracy, standard deviation, and weight.

Learner	Accuracy	Std. dev.	Weight
Boosting	0.55	0.04	9.92
Decision Tree	0.55	0.04	9.89
Naïve Bayes	0.62	0.04	11.55
Nearest Neighbour	0.51	0.03	12.80
Neural Network	0.62	0.01	19.47
Random Forest	0.58	0.04	10.20
Regularized Multinomial	0.63	0.04	10.02
Support Vector Machine	0.52	0.03	12.67
Multinomial	0.40	0.09	3.71

heading 'weight' displays random-effects accuracy weights: the larger the weight, the more precise the estimate of the accuracy.⁵

The Neural Network presents the largest weight (19.47), the worst one being obtained by the Decision Tree (9.89). Overall, the behaviour of the Neural Network in this dataset is good, as it performs well both in terms of accuracy and precision.

Finally, the row with heading 'Overall' sets out the expected accuracy performance obtained by combining the accuracies of all learners via a weighted average with weights equal to the ones reported in the column 'Weight' of Table 2. The result can be interpreted as the expected accuracy and standard deviation of an ensemble learner taking the form, as usual in classification settings, of a *majority vote* classifier. Such classifier would reach an accuracy of 57% with 95% confidence interval ranging between 53% and 61%, smaller than the ones provided by all the other learners, excluding the Neural Network. This result was largely expected as ensemble learners are purposely built for reducing estimation uncertainty, generally at negligible expenses in terms of accuracy reduction. Observe that, also by excluding the standard multinomial accuracy

⁵We assume that the observed learner-specific accuracies $\hat{\theta}_j$, $j = 1, \dots, M$, represent a random sample from a population that is normally distributed with mean θ and variance τ^2 . The weights are thus obtained considering a random-effects model where the $\hat{\theta}_j = \theta_j + \epsilon_j = \theta + u_j + \epsilon_j$ where ϵ_j and u_j are assumed to be independent with $\epsilon_j \sim N(0, \hat{\sigma}_j^2)$ and $u_j \sim N(0, \tau^2)$. The weights are thus calculated as $\hat{w}_j = 1/(\hat{\sigma}_j^2 + \hat{\tau}^2)$, with $\hat{\sigma}_j^2$ obtained by cross-validation, and $\hat{\tau}^2$ by the random-effects maximum likelihood.

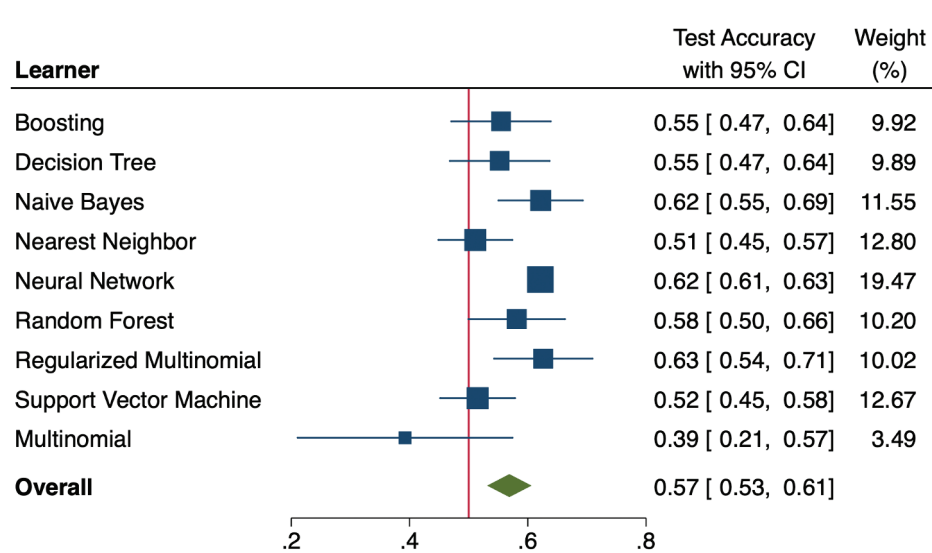


Figure 3. Plot of learners' predictive accuracy, standard deviation, and weight. The dependent variable, wage, is categorical.

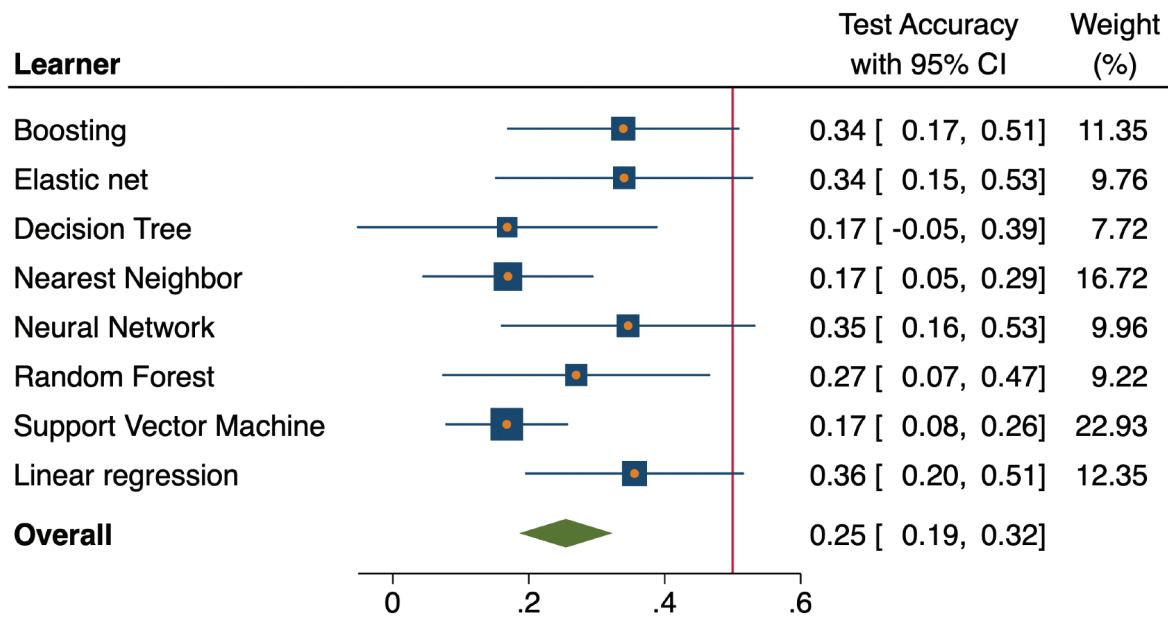


Figure 4. Plot of learners' predictive accuracy, standard deviation, and weight. The dependent variable, wage, is continuous.

from the count, the ensemble accuracy and standard deviation would change slightly.

For the sake of completeness, we also consider wage as continuous instead of categorical variable and run a series of machine learning regressions (instead of classifications) including a standard linear regression. Figure 4 sets out that, in this specific dataset, ML methods do not outperform linear

regression and that all methods have poor predictive power (with an average cross-validated R-squared of around 0.25) and high imprecision.

IV. Conclusion

As part of Artificial Intelligence (AI), Machine Learning is revolutionizing the way we do science.

Instead of relying on hard-wired analytical procedures, the foundational underpinnings of Machine Learning are statistical and computational, fostering scientific breakthroughs on the basis of data-driven predictive performance.

In this spirit, our study has proposed an ML toolbox for improving prediction of targeted outcomes by model's *optimal tuning* and comparing (combining) results from various predictive algorithms available in the literature.

Of course, the benefit of exploring a larger menu of highly predictive statistical procedures can be offset by the non-negligible computational cost that these methods can sometimes require. The growing computational power of computing machines, as well as advances in algorithms' efficiency, can reduce this cost, thus making ML an increasingly appealing tool for applied social scientists.

As a limitation, this article did not directly investigate the link between data configuration and optimal ML predicting algorithms. For this purpose, appropriate Monte Carlo experiments can be designed. By artificially producing a variety of different data generating processes (and thus datasets) characterized by different degrees of sparseness, classes' size, imbalances, and so on, one could discover which ML algorithms perform better according to specific data configurations. This could be the basis for a brand-new study that could complement the one proposed in this article.

Disclosure statement

No potential conflict of interest was reported by the author.

ORCID

Giovanni Cerulli  <http://orcid.org/0000-0002-5892-3372>

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